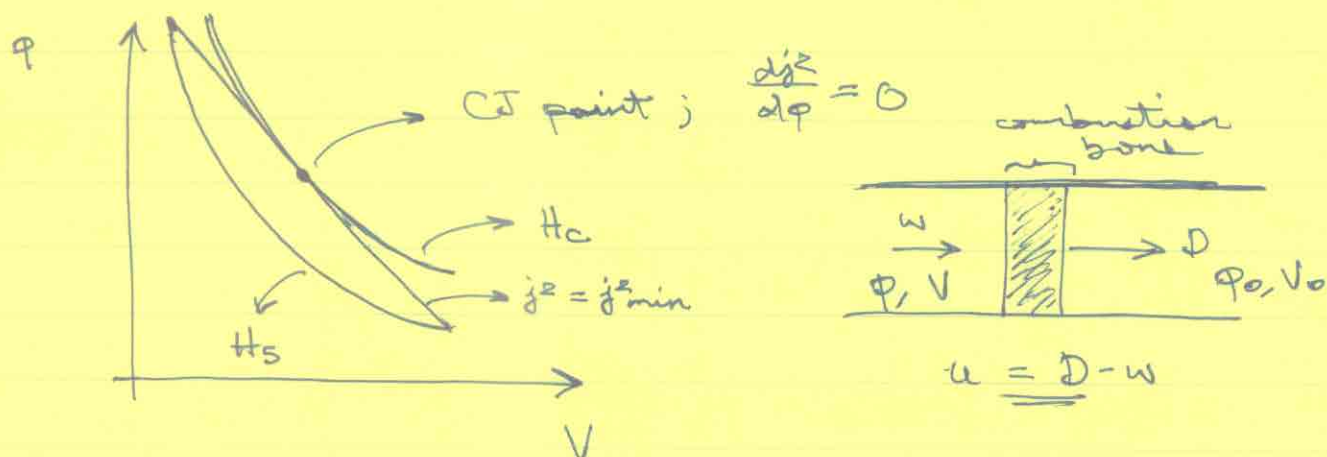


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①

Conditions at the CJ Point (Fickett + Davis)



$$j^2 = \frac{D^2}{V^2} = \frac{p - p_0}{V_0 - V} \Rightarrow V = V_0 - \frac{V_0^2}{D^2} (p - p_0)$$

$$\Rightarrow \underline{\underline{\frac{V}{V_0} = 1 - \frac{V_0}{D^2} (p - p_0)}}$$

$$\text{@ CJ, } u = c \Rightarrow \frac{u^2}{V^2} = j^2 = \frac{p - p_0}{V_0 - V} = \frac{c^2}{V^2} = - \frac{\partial p}{\partial V}|_s$$

$$\gamma = - \frac{\partial \ln p}{\partial \ln V}|_s = - \frac{V}{p} \frac{\partial p}{\partial V}|_s = \frac{V}{p} \frac{p - p_0}{V_0 - V}$$

$$\Rightarrow \gamma = \frac{1 - p_0/p}{V_0/V - 1} \Rightarrow \boxed{\frac{V_0}{V} = \frac{\gamma + 1 - p_0/p}{\gamma}}$$

$$\Rightarrow 1 - \frac{V_0}{D^2} \Delta p = \frac{\gamma}{\gamma + 1 - p_0/p}$$

$$\Rightarrow 1 - \frac{\gamma}{\gamma + 1 - p_0/p} = \frac{1 - p_0/p}{\gamma + 1 - p_0/p} = \frac{\Delta p/p}{\gamma + 1 - p_0/p} = \frac{V_0}{D^2} \Delta p$$

$$\Rightarrow \boxed{p = \frac{D^2}{V_0} (\gamma + 1 - p_0/p)^{-1}}$$

$$u = c \Rightarrow w = D - c ; \quad u^2 = c^2 = \gamma p V$$

$$\begin{aligned} \delta \phi V &= \delta \left\{ \frac{b^2}{V_0} (\delta + 1 - \frac{p_0}{p})^{-1} \right\} \left\{ V_0 (\delta + 1 - \frac{p_0}{p})^{-1} \right\} \\ &= \delta^2 b^2 (\delta + 1 - \frac{p_0}{p})^{-2} \end{aligned}$$

$$\Rightarrow u = D \delta (\delta + 1 - \frac{p_0}{p})^{-1}$$

$$\begin{aligned} D - u &= D - c = D \left[\frac{\delta + 1 - \frac{p_0}{p} - \delta}{\delta + 1 - \frac{p_0}{p}} \right] \\ &= \boxed{D \left[\frac{1 - \frac{p_0}{p}}{\delta + 1 - \frac{p_0}{p}} \right]} = \underline{\underline{w}} \end{aligned}$$

For an ideal gas, the H relation gives

$$c_p T - c_{p0} T_0 - q = \frac{1}{2} (p - p_0) (V_0 + V) .$$

Let $q \gg c_{p0} T_0, p_0 V_0$

$$\Rightarrow c_p T - q \approx \frac{1}{2} p (V_0 + V) . \quad (\text{approx. H curve})$$

as $V \rightarrow V_0$, we find

$$p V_0 \approx \frac{1}{\delta - 1} p V_0 - q \Rightarrow \frac{1}{\delta - 1} p V_0 \approx q$$

$$\Rightarrow \underline{\underline{\phi \approx (\delta - 1) q / V_0}} .$$

$$\text{If } q \gg p_0 V_0 \Rightarrow p/p_0 \gg 1 \text{ at } V \rightarrow V_0 .$$

since $\phi(V < V_0) > \phi(V_0)$, we have $P_{\phi_0} \gg 1$ over all of H .

If we neglect $\phi \gg p$ in the above relations:

$$\boxed{\begin{aligned} \phi &= \frac{1}{\delta+1} \frac{D^2}{V_0} \\ \frac{V}{V_0} &= \frac{\delta}{\delta+1} \rightarrow \text{maximum compression} \\ w &= \frac{1}{\delta+1} D \end{aligned}}$$

Given (D, V_0) we obtain (ϕ, V, w) .

Now, if we assume the approx. H curve:

$$\frac{\delta}{\delta+1} \phi V - q \approx \phi(V + V_0)$$

$$\Rightarrow \phi \left[\frac{\delta}{\delta+1} \frac{V}{V_0} - \frac{1}{2} \left(1 + \frac{V}{V_0} \right) \right] V_0 \approx q$$

$$\Rightarrow \underline{\underline{p_0 \frac{V_0}{2} \left[\left(\frac{\delta+1}{\delta-1} \right) \frac{V}{V_0} - 1 \right] \approx q}}$$

$$\text{Let } \frac{V}{V_0} = \frac{\delta}{\delta+1} \Rightarrow p \frac{V_0}{2} \left[\frac{\delta}{\delta-1} - 1 \right] \approx q$$

$$\Rightarrow p \approx \frac{2q}{V_0} (\delta-1) \quad \left\{ \begin{array}{l} \text{twice the value} \\ \text{at } V \rightarrow V_0 \end{array} \right.$$

So, if we know q , then

$$D^2/\chi_0 \cdot \frac{1}{f+1} \approx \frac{\partial q}{\partial \chi_0} (f-1)$$

$$\Rightarrow \boxed{D^2 \approx \partial q (\chi^2 - 1)}.$$

We now have the complete solution for a strong CT detonation wave in terms of the reaction heat q .